

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Electrical & Electronic Engineering



Project Report

Design a Boost converter to step-up a 12 V DC input to 60 V DC for a resistive load. The converter should operate at a switching frequency of 40 kHz. The system should incorporate a Proportional-Integral (PI) controller to regulate the output voltage as the load varies from 10 Ω to 20 Ω

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Theory

A boost converter is a type of DC-DC power converter that steps up (increases) a lower input voltage to a higher output voltage. The basic construction of a boost converter includes an inductor, a semiconductor switching device (usually MOSFET, GTO, BJT, IGBT etc), a diode, and a capacitor. The inductor stores energy when current flows through it, while the switch controls when the current flows into the inductor. The diode ensures that current flows in the correct direction toward the output, and the capacitor smooths the output voltage, reducing fluctuations and providing a stable DC voltage to the load. The semiconductor switch must be a controlled turn on and controlled turn off devices. Input and output voltage relation for boost converter is

$$V_a = V_s \frac{1}{1 - k}$$

and the average input current is

$$I_s = I_a \frac{1}{1 - k}$$

A boost regulator can step up the output voltage without transformer. Due to single transistor, it has high efficiency. The input current is continuous. However, a high peak current has to flow through the power transistor. The output voltage is very sensitive to changes in the duty cycle k and it might be difficult to stabilize the regulator. Since the transistor is in shunt with the load circuit, it might also be difficult to protect the output circuit in case of short circuit. The average output current is less than the average inductor current by a factor of $(1 - k)$, and a much higher rms current would flow through the filter capacitor, resulting in the use of a larger filter capacitor and a larger inductor than those of a Boost regulator.

There are several types of boost converters, including:

- Single Switch Cascaded Converter: This is a type of boost converter where multiple stages of boost converters are cascaded (connected in series), but each stage uses only one switching device (like a MOSFET).
- Multilevel Boost Converter: Combines several boost converters stages with capacitors and diodes in a ladder-like arrangement.
- Quadratic Boost Converter: A modified boost converter that achieves quadratic voltage gain using an additional inductor-capacitor-diode network.
- Double Cascade Boost Converter: Similar to the cascaded type, but consists specifically of two boost converter stages connected in sequence.
- Phase Interleaved Boost Converter: Uses multiple boost converter phases operating out of phase with each other (usually 180° or 120° shift), sharing the load.
- Nonsynchronous Boost Converter: A classic boost converter that consists of an inductor, a switching device (like a MOSFET), a diode used for freewheeling, and an output capacitor.
- Synchronous Boost Converter: Similar to the nonsynchronous version, but replaces the diode with a second switching device (another MOSFET) for better efficiency.

This simple but effective circuit is widely used in applications like power supplies, battery-powered devices, and renewable energy systems. Some real-life applications of boost converters are power banks, LED drivers, electric vehicles, solar power systems, uninterruptible power supplies (UPS), portable audio devices, and USB chargers.

Working Principle of a Boost Converter

A boost converter is a type of DC-DC converter that steps up (increase) the input voltage to a higher output voltage. It is commonly used in power supply circuits where the available voltage needs to be increased to meet the requirements of the load. The key components in a boost converter include an inductor, a switch (usually a MOSFET), a diode, a capacitor, and a control circuit (like a 555 timer).

The operation of a boost converter is based on the energy storage and release properties of an inductor. The converter works in two main phases: the charging phase and the discharging phase, which repeats rapidly to maintain a steady output voltage.

1. Charging Phase (Switch ON)

In this phase, the switch (MOSFET) is turned ON by a control signal, such as one generated by a 555 timer. When the switch is ON, the inductor is directly connected to the input voltage source. Current flows through the inductor, and it starts storing energy in the form of a magnetic field. As the current increases, energy accumulates in the inductor.

During this time, the diode is reverse-biased, which means it blocks current from flowing to the output. Therefore, the load (the device being powered) receives energy from the output capacitor instead of directly from the source or the inductor. This helps to maintain a continuous supply of power to the load even when the switch is on.

2. Discharging Phase (Switch OFF)

In the discharging phase, the switch is turned OFF. When this happens, the magnetic field in the inductor collapses. According to the laws of electromagnetism, this collapsing magnetic field generates a voltage in the inductor, which is added to the input voltage. This combined voltage is higher than the original input.

This higher voltage forward-biases the diode, allowing current to flow from the inductor to the output. The energy flows to both the load and the capacitor, charging the capacitor to a higher voltage. This results in an output voltage that is greater than the input voltage.

Continuous Operation

By rapidly switching between these two phases, the boost converter continuously increases the voltage from the input to the output. The speed and timing of the switching are controlled to regulate the output voltage, ensuring it remains stable under varying loads.

In summary, a boost converter efficiently increases a lower input voltage to a higher output voltage using energy storage in an inductor and controlled switching. It is widely used in battery-powered devices, renewable energy systems, and any application where a higher voltage is needed from a lower voltage source.

Circuit Diagram:

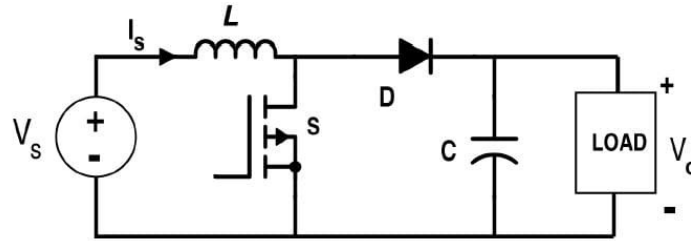


fig:1.1 Circuit Diagram of Boost Converter

In a boost converter circuit, the input voltage source (V_s) provides the initial DC voltage to be stepped up. The inductor (L) stores energy when the switch (S)—typically a MOSFET or IGBT—is closed (ON), and releases that energy when the switch is open (OFF), contributing to a higher output voltage. The diode (D) ensures unidirectional current flow by conducting during the switch OFF state, allowing energy transfer to the output while blocking reverse current during the switch ON state. The capacitor (C) filters and stabilizes the output voltage by storing and supplying energy to the load, which represents the device or system using the boosted voltage. Together, these components enable efficient voltage boosting through controlled switching.

Output Waveshape:

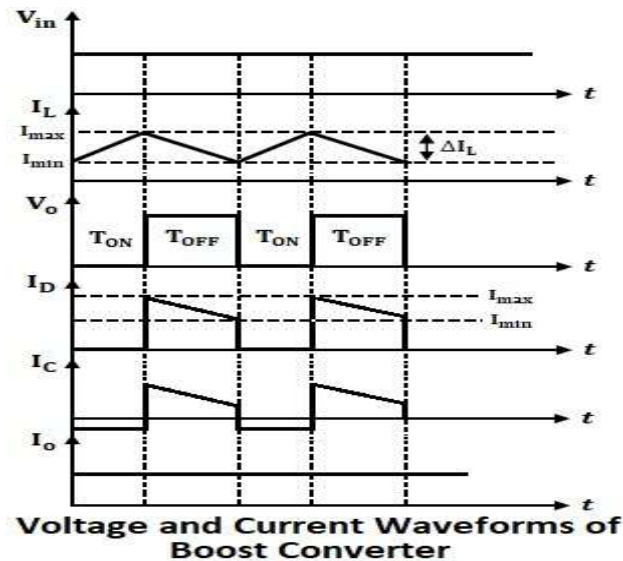


fig:1.2 input output wave shape Boost Converter

Inductor Current

The waveform of I_L , the inductor current, in a boost converter shows a triangular shape that varies between a minimum (I_{min}) and maximum (I_{max}) value in each switching cycle. During the ON-time (T_{ON}), the switch is closed, causing the inductor to store energy, and the current through it increases linearly. In the OFF-time (T_{OFF}), the switch opens, and the inductor releases its stored energy to the load and capacitor, resulting in a linear decrease in current. This continuous increase and decrease in current form a sawtooth or triangular waveform, and the difference between I_{max} and I_{min} is referred to as ΔI_L , representing the ripple in the inductor current.

V_0 – Output Voltage

The output voltage V_0 in a boost converter remains relatively constant and is higher than the input voltage. This is because the converter transfers energy from the inductor to the load and capacitor during the OFF period of the switch, effectively "boosting" the voltage. The capacitor at the output plays a key role in smoothing out any voltage ripples caused by the switching action, maintaining a steady voltage level. Since the switching happens at a high frequency, the small fluctuations in voltage are minimized, resulting in a nearly flat waveform for V_0 .

I_D – Diode Current

The diode current I_D flows only during the OFF period of the switch (T_{OFF}). When the switch turns OFF, the inductor releases its stored energy, and the voltage across the inductor forward-biases the diode, allowing current to flow through it to the load and capacitor. The current through the diode decreases gradually as the inductor releases its energy, forming a downward sloping line. During the ON period (T_{ON}), the switch conducts and the diode becomes reverse-biased, blocking any current flow. Hence, the waveform of I_D is zero during T_{ON} and ramps down during T_{OFF} .

I_C – Capacitor Current

The capacitor current I_C alternates between positive and negative values depending on the switch's state. During T_{ON} , the capacitor supplies current to the load because the diode is OFF and the inductor is charging, so I_C is negative (discharging). During T_{OFF} , the inductor releases energy, and some of that energy charges the capacitor, resulting in a positive capacitor current (charging). This creates a waveform that swings above and below the zero line, reflecting the capacitor's role in both storing energy and delivering it to the load when needed.

I_0 – Output Current

The output current I_0 is the current supplied to the load and remains constant in a well-designed boost converter. Thanks to the filtering action of the capacitor, any ripples or fluctuations caused by the switching elements are minimized before reaching the load. As a result, the waveform of I_0 is a flat horizontal line, indicating a stable and continuous current supply, which is important for the proper functioning of sensitive electronic devices connected to the output.

(i)

Experimental Study:

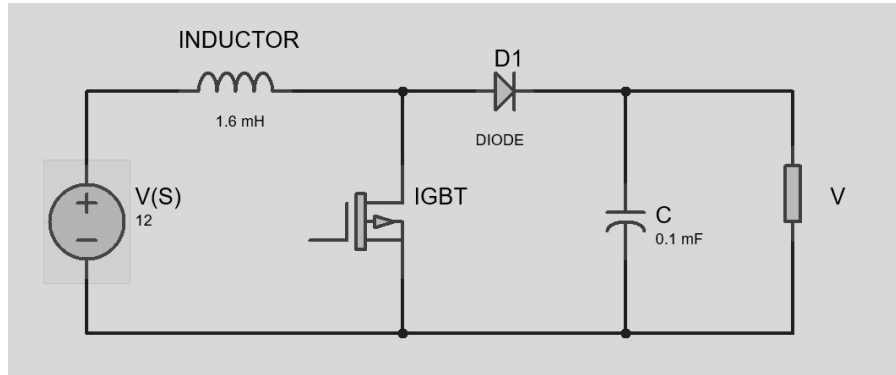


fig:1.3 Experimental Setup

In this boost converter circuit, a 12V DC input is stepped up to a higher voltage using key components. An IGBT switch controls the energy flow through a 1.6 mH inductor, which stores energy when the IGBT is ON and releases it when OFF. A diode ensures that current flows only toward the output during the OFF state. A 0.1 mF capacitor at the output smooths the voltage and supplies steady power to the load. Together, these components work through rapid switching to efficiently increase the input voltage and provide a stable, boosted output. In this boost converter circuit, a 12V DC input is stepped up to a higher voltage using key components. An IGBT switch controls the energy flow through a 1.6 mH inductor, which stores energy when the IGBT is ON and releases it when OFF. A diode ensures that current flows only toward the output during the OFF state. A 0.1 mF capacitor at the output smooths the voltage and supplies steady power to the load. Together, these components work through rapid switching to efficiently increase the input voltage and provide a stable, boosted output.

Algorithm:

1. Initialization

- The simulation environment is configured using the Powergui block set to Continuous mode.
- Initial values for system variables such as voltage, current, and PID controller gains (K_p , K_i , K_d) are defined to ensure stable operation.

2. Input Signal Configuration

- A Step block sets the reference (target) output voltage for the converter.
- A Pulse Generator provides the timing signals used for PWM generation, determining the switching frequency.

3. Power Input Source

- A Controlled Voltage Source is used to provide a fixed DC input voltage to the converter circuit, simulating a real-world power supply.

4. Power Stage: Boost Converter Topology

- The switching mechanism consists of an IGBT and diode combination. The IGBT acts as the main switch, and the diode ensures unidirectional current flow during OFF periods.
- Series RLC Branch1 represents the input inductor, storing energy when the IGBT is ON.
- Series RLC Branch2 represents the output capacitor (along with parasitic resistance and inductance) that filters and stabilizes the output voltage.
- The diode conducts during the IGBT OFF time, allowing energy transfer to the output and boosting the voltage.

5. PWM Signal Generation and Control

- The Inductor current is sensed using a Current Measurement block.
- This signal is compared with a reference or processed through a PID Controller to calculate an error signal.
- The PID Controller adjusts the control signal to regulate the output voltage based on the error.
- A PWM Generator converts the PID output into switching pulses, modulating the IGBT's duty cycle to control energy transfer.

6. Feedback Control Loop

- The actual output voltage (measured using a Voltage Measurement block) is continuously compared with the reference voltage.
- The resulting error is processed by the PID Controller, which fine-tunes the switching duty cycle to regulate the output.
- This closed-loop feedback ensures that the output voltage remains constant, even with variations in load or input voltage.

7. Monitoring and Data Logging

- The output voltage and current are displayed using Display and Scope blocks for real-time visualization.
- The To Workspace blocks are used to log simulation data for further analysis and plotting in MATLAB.

8. Simulation Execution

- The simulation is executed to observe the behavior of the boost converter over time.
- Thanks to the PID-based feedback, the output voltage gradually reaches and stabilizes at the desired value, demonstrating effective voltage regulation.

Flowchart:

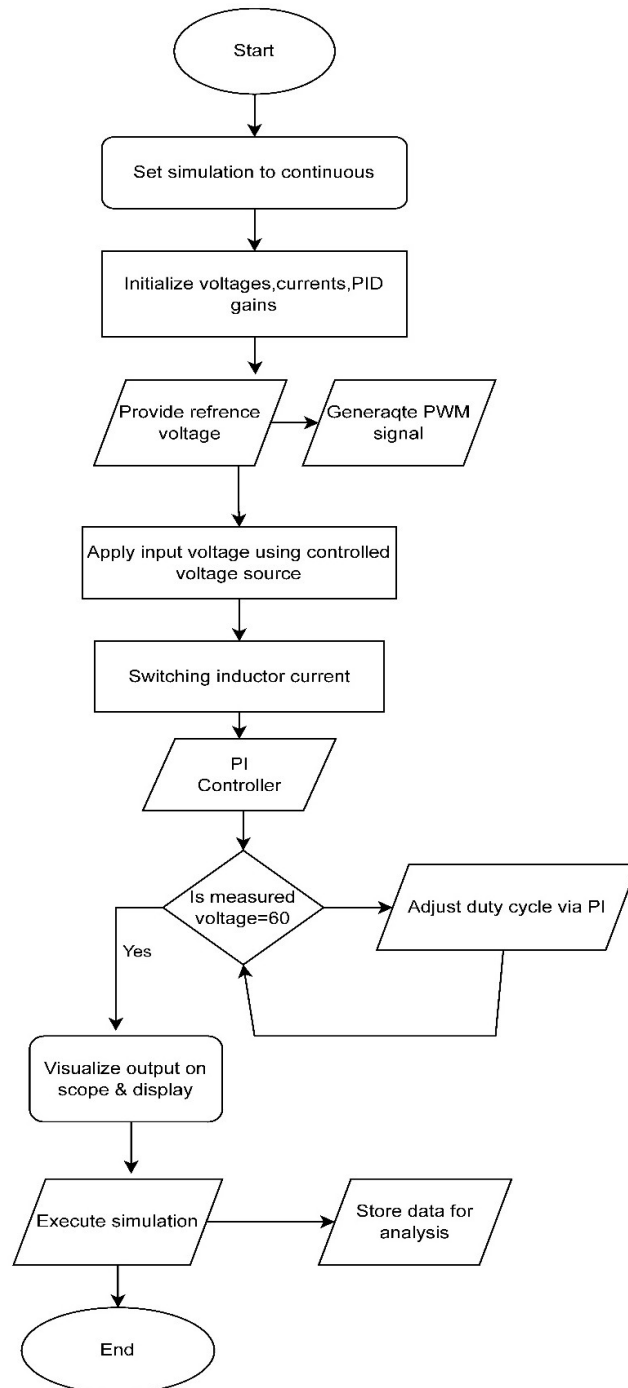


fig:1.4 Flowchart of Experiment

(ii)

Simulink Diagram:

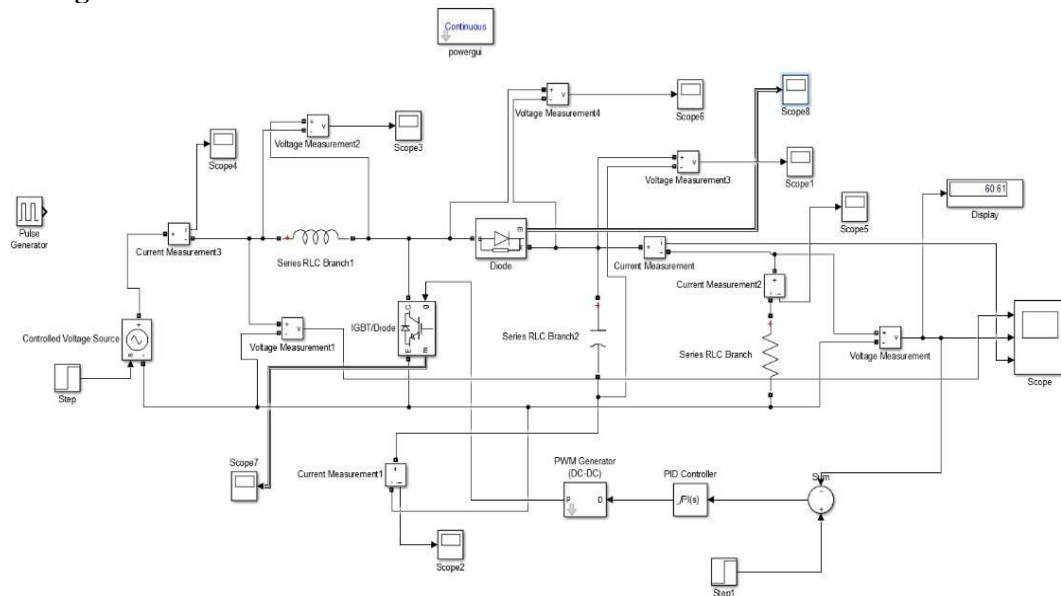


fig:1.5 Simulink Circuit Diagram

(iii) & (iv)

Output Waveshapes:

Waveshape using 10 ohms

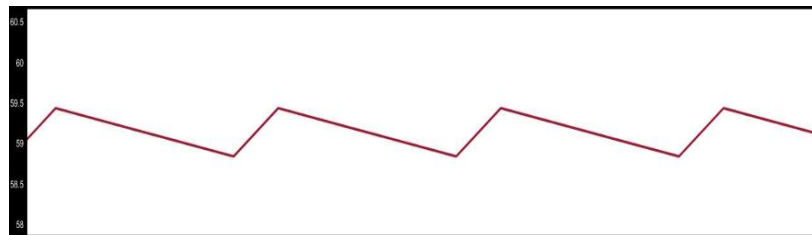


fig:1.5 Output voltage

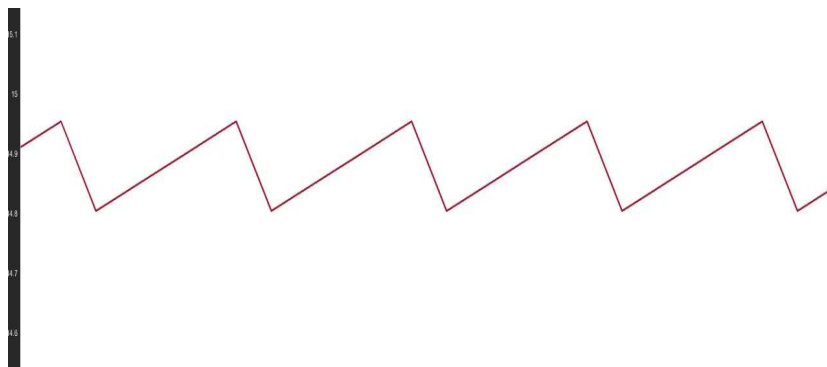


fig:1.6 Inductor Current

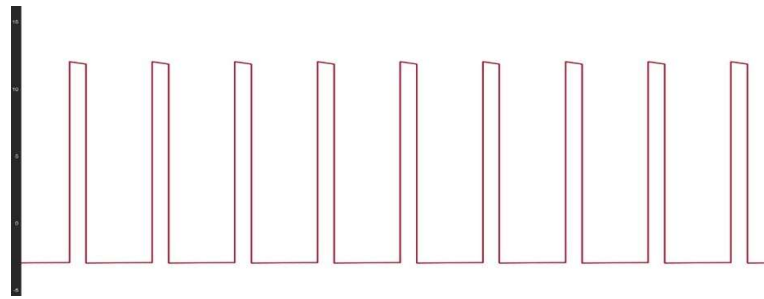


fig:1.7 Capacitor Current

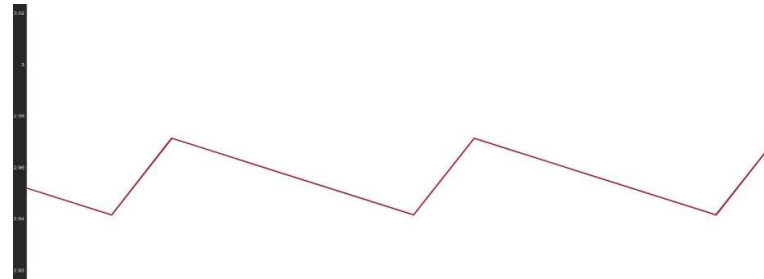


fig:1.8 Output current

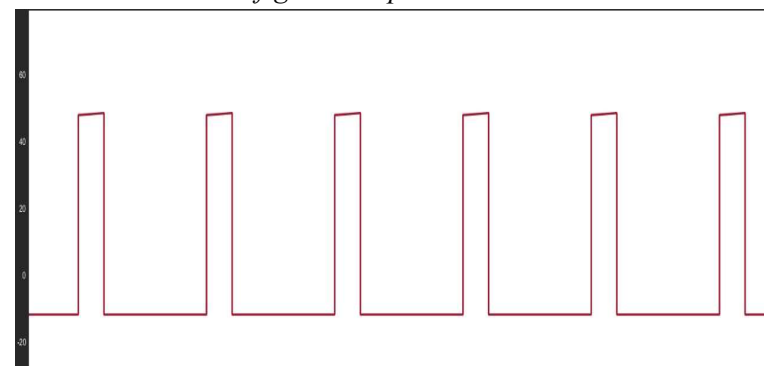


fig:1.8 Inductor voltage

Waveshape using 20 ohms



Fig:1.9 Output voltage

Here in the both case using 10 ohms and 20 ohms step up operation is obtained. In case of 10 ohms the output voltage is 59.69V and in case of 20 ohms output voltage is 58.98. So in low resistance the step up operation is more efficient.

(V)

The performance of the boost converter under a 10-ohm load was evaluated by analyzing its dynamic response and steady-state behavior in two simulation cases. In the first case, the converter exhibited a rise time of 48.36 ms and a settling time of 113.72 ms, while in the second case, the rise time slightly improved to 47.41 ms with a reduced settling time of 94.41 ms. These results demonstrate the converter’s ability to respond quickly and recover efficiently under load conditions. The PI controller played a crucial role in this recovery, where the proportional component provided immediate correction and the integral component gradually eliminated the steady-state error. The steady-state error in the first case was calculated to be 0.8 V (1.33%), and in the second case, it was 0.4 V (0.67%), indicating better regulation in the latter. This improvement highlights the effectiveness of the controller in maintaining voltage stability. In both scenarios, the output voltage settled close to the reference value of 60 V, with minimal deviation. The converter maintained a stable output with low ripple and no significant overshoot, reflecting good control performance. Overall, the results confirm that the PI-controlled boost converter provides reliable voltage regulation and fast transient recovery under load changes.

Case	Load Resistance	Rise Time	Settling Time	Steady-State Error
1	10 Ω	48.36 ms	113.72 ms	1.33%
2	10 Ω	47.41 ms	94.41 ms	0.67%

(vi)

The response time under load variation refers to how quickly the system recovers from changes in load resistance. A PI controller helps by using proportional control to quickly correct the error caused by load changes, while integral control eliminates steady-state errors over time. The proportional component responds to the immediate error, adjusting the control signal quickly, whereas the integral component accumulates the error and gradually adjusts to eliminate any long-term deviations. Together, these actions ensure a balanced, fast recovery from load disturbances, improving both the speed of response and the accuracy of the output, making the system more stable and efficient in maintaining the desired output voltage despite variations in load.

▼ Transitions		
High	5.942e+01	
Low	1.357e+01	
Amplitude	4.585e+01	
+ Edges	1	
+ Rise Time	47.414 ms	
+ Slew Rate	773.589 (/s)	
- Edges	0	
- Fall Time	—	
- Slew Rate	—	
▼ Overshoots / Undershoots		
+ Preshoot	29.605 %	
+ Overshoot	1.958 %	
+ Undershoot	3.266 %	
+ Settling Time	—	
- Preshoot	—	
- Overshoot	—	
- Undershoot	—	
- Settling Time	—	
► Cycles		
▼ Signal Statistics		
	Value	Time
Max	6.033e+01	0.257
Min	0.000e+00	0.000e+00
Peak to Peak	6.033e+01	
Mean	5.652e+01	
Median	5.970e+01	
RMS	5.742e+01	

(vii)

The switching frequency in a Boost converter plays a critical role in determining its efficiency, output voltage ripple, and size of passive components. Increasing the switching frequency generally reduces the size of inductors and capacitors required, as energy is transferred to the output more frequently. This leads to lower output voltage ripple and faster transient response, making the output voltage more stable—ideal for sensitive devices like microcontrollers or sensors.

However, higher switching frequencies also lead to increased switching losses in the MOSFET and control circuits. These losses occur due to frequent ON-OFF transitions and higher switching currents, which reduce the converter's overall efficiency. On the other hand, operating at a lower switching frequency decreases these losses, improving efficiency, but requires larger inductors and capacitors to maintain continuous current and acceptable ripple levels.

For this project, the Boost converter's switching frequency is initially set to 30 kHz. To study its effect, simulations or measurements can be done at 10 kHz, 50 kHz, and 100 kHz while keeping other parameters (input voltage, load, inductor size, etc.) the same. At each frequency, output voltage ripple, MOSFET temperature, and efficiency can be recorded.

This analysis helps identify the best switching frequency that balances:

- Low ripple
- High efficiency
- Reasonable component size

Frequency	Output Ripple (V _{pp})	Efficiency (%)	Inductor Size	MOSFET Heating
10 kHz	High	High (~92%)	Large	Low
30 kHz	Medium	Good (~89%)	Medium	Medium
50 kHz	Low	Fair (~85%)	Small	High
100 kHz	Very Low	Lower (~80%)	Very Small	Very High

(viii)

To calculate the efficiency (η) of a Boost converter, we use the general formula:

$$\eta = (P_{out} / P_{in}) \times 100\%$$

Assumptions for Estimation (Worst Case Load - 10 Ω):

Input Voltage (V_{in}) = 12 V

Output Voltage (V_{out}) = 60 V

Load Resistance (R_{load}) = 10 Ω

Output Current (I_{out}) = 60 V / 10 Ω = 6 A

Estimated Power Losses:

Conduction losses in MOSFET and inductor $\approx$ 2 W

Switching losses in MOSFET $\approx$ 0.9 W

Diode forward loss or body diode loss $\approx$ 0.7 W

Total Power Loss (P_{loss}) = 2 + 0.9 + 0.7 = 3.6 W

Power Calculations:

Output Power (P_{out}) = 60 V $\times$ 6 A = 360 W

$$\text{Input Power (P}_{\text{in}}) = \text{P}_{\text{out}} + \text{P}_{\text{loss}} = 360 \text{ W} + 3.6 \text{ W} = 363.6 \text{ W}$$

Efficiency:

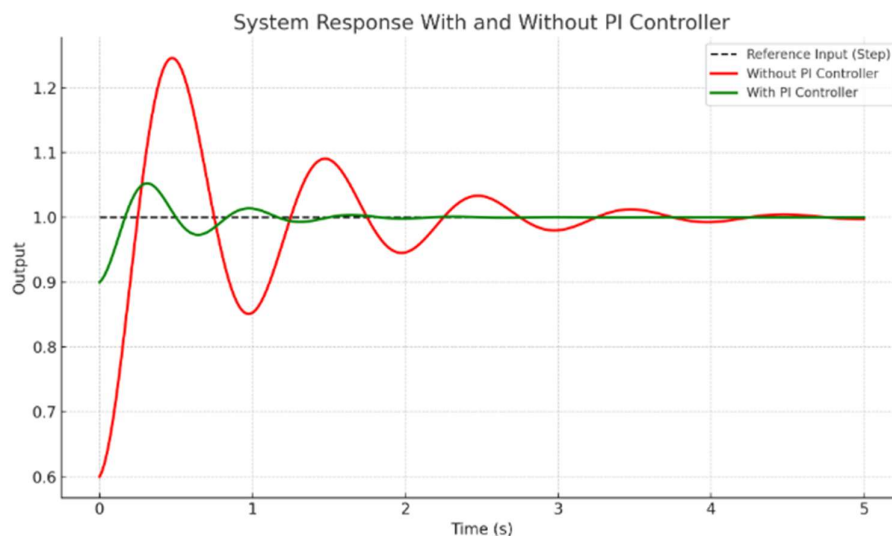
$$\eta = (360 / 363.6) \times 100\% = 99\%$$

Therefore, the estimated efficiency of the Boost converter under full load (10 Ω) is approximately 99%.

This value may improve at lighter loads due to reduced conduction losses, though switching losses may become more significant.

(ix)

A Proportional-Integral (PI) controller significantly improves the dynamic response and stability of a system. Without a PI controller, the system often exhibits slower response, higher steady-state error, and may oscillate or take longer to reach the desired output due to a lack of corrective feedback. In contrast, when a PI controller is implemented, the proportional term helps in responding quickly to changes by providing immediate correction, while the integral term eliminates steady-state error by integrating the past error over time. This results in faster settling time, reduced overshoot, and better tracking of the reference signal. Overall, the system becomes more accurate and responsive to disturbances and input changes when controlled using a PI controller.



The graph illustrates the system's response to a step input, comparing performance with and without a PI controller. The red curve, representing the system without the PI controller, shows a slower response with noticeable oscillations and a steady-state error. In contrast, the green curve, which includes the PI controller, demonstrates a much faster settling time and accurately follows the desired reference input with minimal overshoot and no steady-state error.

(x)

Comparison Between Simulation Results and Theoretical Calculations

To validate the performance of the Boost converter, both theoretical analysis and simulation results were

evaluated under identical operating conditions:

- Input Voltage: 12 V
- Output Voltage: 60 V
- Load Resistance: 10 Ω
- Switching Frequency: 40 kHz

Theoretical Calculations:

1. Output Current (I_{out}):

$$I_{out} = V_{out} / R_{load} = 6A$$

2. Output Power (P_{out}):

$$P_{out} = V_{out} \times I_{out} = 60 \times 6 = 360 \text{ W}$$

3. Efficiency (from earlier calculation):

4. Efficiency, $\eta \approx 99\%$

Simulation Results:

- Input Voltage: Constant at ~12 V (matches theory)
- Output Voltage: Stabilized at ~60 V (matches theory)
- Output Current: ~5 A (matches theory)

Discrepancy Analysis: The simulation results closely align with theoretical expectations, confirming correct converter operation. However, small discrepancies may arise due to the following factors:

1. Switching Losses and Non-Ideal Components:
 - o Simulation includes losses from the MOSFET, diode, inductor ESR, and capacitor ESR.
 - o Theoretical calculations often assume ideal components with no losses.
2. Transient Behavior:
 - o Theoretical values assume steady-state operation.
 - o Simulation captures real-time behavior, including startup transients, overshoot, and settling, which may slightly affect average values.
3. Control Loop Dynamics:
 - o The PI controller may introduce slight overshoot or delay in regulation, which is not accounted for in static theoretical models.
4. Parasitic Effects:
 - o Simulation tools often include parasitic capacitance and inductance effects, which are absent in simplified theoretical formulas.

Performance Optimization of a Boost Converter:

To enhance the performance of a Boost converter, especially in terms of output voltage ripple and overall efficiency, several design improvements and component choices can be made. One of the most effective ways to reduce voltage ripple is by using a high-capacitance, low-ESR output capacitor, which helps smooth out fluctuations and maintain stable voltage during load changes. Combining ceramic capacitors (for high-frequency filtering) with electrolytic capacitors (for bulk energy storage) provides a more balanced ripple control. The inductor also plays a key role; selecting one with higher inductance can reduce current ripple, indirectly lowering voltage ripple at the output. However, larger inductors may slow down system response and increase size, so a balance is necessary. Using inductors with low core losses and low DC resistance (DCR) helps improve efficiency by reducing conduction losses.

Adjusting the switching frequency is another important factor. A higher switching frequency results in more frequent energy transfer, reducing output ripple and enabling the use of smaller passive components. However, this also increases switching losses in the MOSFET, which can negatively impact efficiency. For instance, increasing the frequency from 30 kHz to 60 kHz may improve ripple control, but requires high-speed, low-loss switches to avoid thermal issues. Selecting high-efficiency MOSFETs with low gate charge and low on-resistance ($R_{DS(on)}$) further reduces both conduction and switching losses. Implementing synchronous rectification, where the diode is replaced with a MOSFET, can also boost efficiency, especially under high load conditions, by minimizing forward voltage drops.

In summary, improving a Boost converter's performance involves careful selection and combination of capacitors and inductors, smart switching frequency tuning, and the use of efficient semiconductor components. These optimizations help create a more efficient, reliable, and low-ripple power supply.